CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 30-2021: A 47-Year-Old Man with Recurrent Unilateral Head and Neck Pain

Mark E. Lindsay, M.D., Ph.D., David M. Dudzinski, M.D.,
Byung C. Yoon, M.D., Ph.D., Michael R. Jaff, D.O., and Joseph V. Thakuria, M.D.

PRESENTATION OF CASE

Dr. David M. Dudzinski: A 47-year-old right-handed man was evaluated at this hospital because of pain on the left side of the head and neck.

Six years before the current evaluation, the patient was admitted to this hospital with 1 week of headache, pain on the left side of the neck, numbness on the left side of the face, and dizziness. The pain was not relieved with the use of ibuprofen and had previously been evaluated and treated by a chiropractor. A neurologic examination revealed a slight decrease in taste and in sensation to touch and temperature on the left side of the face; a gait evaluation revealed a subtle leftward sway. Imaging studies were obtained.

Dr. Byung C. Yoon: Computed tomography (CT) of the head, performed after the administration of intravenous contrast material, revealed subtle foci of hypodensity in the left cerebellum, a finding suggestive of infarction. There was no evidence of arterial dissection, aneurysm, stenosis, or thrombotic occlusion. Magnetic resonance imaging (MRI) of the head, performed before and after the administration of intravenous contrast material, confirmed the presence of a left medial cerebellar infarct and a small left lateral medullary infarct (Fig. 1A, 1B, and 1C).

Dr. Dudzinski: The patient had normal results on cerebrospinal fluid (CSF) analysis, echocardiography, and a hypercoagulability panel, except for a mildly elevated level of anticardiolipin IgM (18.9 IgM phospholipid units; reference range, 0 to 15). During the hospital admission, lesions with a vesicular appearance developed on the left side of the lips and oral cavity, and the patient was treated with famciclovir. He was discharged home while receiving aspirin, with a follow-up visit scheduled in the neurology clinic.

One month later in the neurology clinic, the patient reported partial resolution of headache, persistent neck pain and ear pain with balance difficulty, and perioral paresthesia. There was diminished sensation on the left side of the face (similar to that observed during the previous examination). No oropharyngeal lesions were visible. An evaluation of gait and balance was normal.

Dr. Yoon: Magnetic resonance angiography of the head and neck, performed

From the Departments of Medicine (M.E.L., D.M.D.), Pediatrics (M.E.L., J.V.T.), Radiology (B.C.Y.), and Surgery (M.R.J.), Massachusetts General Hospital, and the Departments of Medicine (M.E.L., D.M.D., M.R.J.), Pediatrics (M.E.L., J.V.T.), and Radiology (B.C.Y.), Harvard Medical School — both in Boston.

N Engl J Med 2021;385:1317-25.
DOI: 10.1056/NEJMcpc2107347

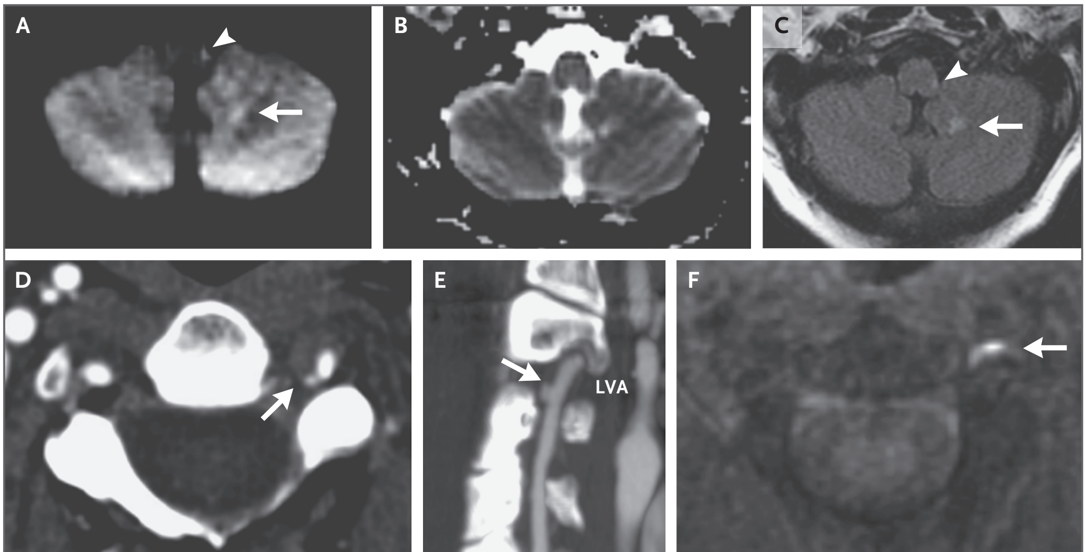

Figure 1. MRI and CT Angiogram.

Six years before the current evaluation, MRI of the head was performed after the administration of intravenous contrast material. A diffusion-weighted image (Panel A) shows subtle foci of hyperintensity in the left medial cerebellum (arrow) and left lateral medulla (arrowhead). On an apparent diffusion coefficient map (Panel B), the corresponding lesions are isointense to hyperintense. On a T2-weighted fluid-attenuated inversion recovery image (Panel C), the lesions are hyperintense (arrow and arrowhead). These findings are consistent with evolving nonacute infarcts. On the current evaluation, CT angiography of the neck was performed after the administration of contrast material. An axial image (Panel D) shows asymmetric irregularity of the left vertebral artery at the C2–C3 level (arrow), a finding consistent with dissection. A maximum-intensity-projection reformation (Panel E) shows a medially directed 2-mm pseudoaneurysm (arrow) along the dissected left vertebral artery (LVA). In addition, MRI was performed before the administration of contrast material. A T1-weighted image obtained at the C2–C3 level (Panel F) shows a curvilinear mural hematoma with intrinsic hyperintensity (arrow) along the dissection site.

before and after the administration of intravenous contrast material, revealed expected evolution of the left cerebellar infarct and no new infarction. The left posterior inferior cerebellar artery was not visualized, possibly because of occlusion or slow blood flow.

Dr. Dudzinski: Repeat CSF analysis showed normal protein and glucose levels and 14 white cells per microliter (reference range, 0 to 5), of which 94% were lymphocytes. Gram's staining showed no organisms. CSF tests for borrelia and varicella DNA and for cytomegalovirus and herpes simplex virus antibodies were negative. Two days later, cerebral angiography revealed mild narrowing of the origin of the left vertebral artery. Tests for antinuclear antibodies, antineutrophil cytoplasmic antibodies, and anti-La and anti-Ro antibodies were negative, as was a rapid plasma reagin test.

The patient received follow-up care at another hospital until the current evaluation, when he presented to the emergency department of this hospital with 3 days of severe pain on exertion that affected the left side of the neck and left temporo-occipital region. The symptoms were similar to those he had reported in the past and were not relieved with the use of ibuprofen or marijuana. The head pain was described as pulsating and pounding, and it worsened when he was in the supine position.

The patient reported right flank pain; the onset of this pain had occurred with lifting of a heavy object and had just preceded the onset of headache. He also reported a tendency to bruise. The review of systems was otherwise normal. His medical history was notable for dyslipidemia. He had undergone repair of meniscus tears in both knees and repair of a traumatic dislocation of the right shoulder. Medications included rosuvastatin and gemfibrozil, as well as acetaminophen as needed. There were no known adverse reactions to medications. The patient was a self-employed laborer and lived with his wife and children in a suburb of Boston. He smoked cigars

Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	On Evaluation, This Hospital
Hemoglobin (g/dl)	13.5–17.5	13.7
Hematocrit (%)	41.0–53.0	38.4
White-cell count (per μl)	4500–11,000	6800
Platelet count (per μl)	150,000–400,000	197,000
Erythrocyte sedimentation rate (mm/hr)	0–11	19
Anticardiolipin IgG (IgG phospholipid units)	0–15	3.8
Anticardiolipin IgM (IgM phospholipid units)	0–15	19.5
Cholesterol (mg/dl)		
Total	<200	196
Low-density lipoprotein	<130	129
High-density lipoprotein	35–100	36
Triglycerides (mg/dl)	40–150	155
Urine toxicology screen	Negative for amphetamines, barbiturates, benzodiazepines, cannabinoids, cocaine, opiates, and phencyclidine	Positive for acetaminophen and cannabinoids

* To convert the values for cholesterol to millimoles per liter, multiply by 0.02586. To convert the values for triglycerides to millimoles per liter, multiply by 0.01129.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

and marijuana occasionally, and he drank up to eight beers per day on weekends. His family history was notable for hypertension and dyslipidemia in both of his parents, as well as early-onset dementia in his father.

The temperature was 36.7°C, the blood pressure 176/119 mm Hg in both arms, the heart rate 72 beats per minute, and the oxygen saturation 98% while the patient was breathing ambient air. A neurologic examination revealed a mild decrease in coordination in the left hand but was otherwise normal. Blood levels of electrolytes, troponin T, C-reactive protein, lipoprotein(a), and glycated hemoglobin were normal, as were the prothrombin time, partial-thromboplastin time, and results of renal function tests. Urinalysis was also normal; other laboratory test results are shown in Table 1.

Dr. Yoon: CT angiography of the head and neck, performed after the administration of intravenous contrast material, revealed dissection of the left vertebral artery, extending from the origin of the artery to the proximal V3 (upper cervical) segment, with a 2-mm pseudoaneurysm at the C2–C3 level (Fig. 1D and 1E). There was no evidence of acute ischemia. MRI of the head, performed before and after the administration of intravenous contrast material, did not reveal any evidence of infarction or a perfusion abnormality. However, a T1-weighted image showed a focus of intrinsic hyperintensity along the left vertebral artery, a finding consistent with hemorrhage within the vessel wall from the dissection (Fig. 1F).

Dr. Dudzinski: The patient was admitted to the neurology service. Control of the blood pressure required multiple medications. Echocardiography showed no cardiac, valvular, or aortic structural abnormality. Right flank pain recurred. Urinalysis showed 5 to 10 red cells per high-power field but was otherwise normal.

Dr. Yoon: CT of the abdomen and pelvis, performed after the administration of intravenous contrast material, revealed dissection of the right renal artery, extending from the origin of the artery to subsegmental renal arteries. Perinephric fat stranding was present on the right side, and there were areas of diminished enhancement in

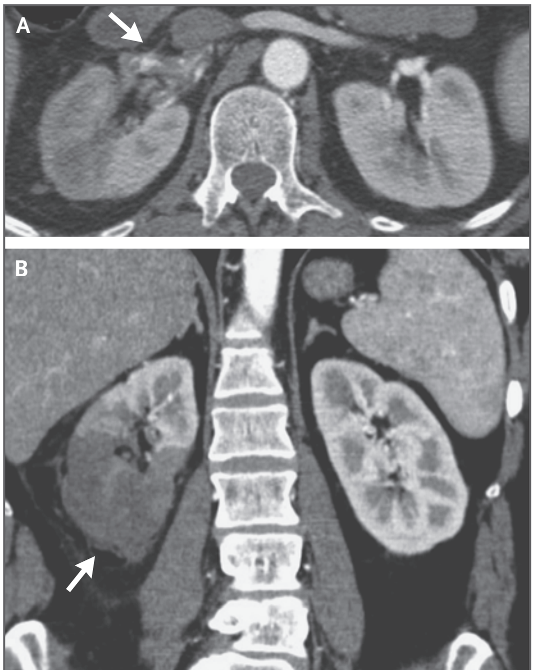

Figure 2. CT Scan of the Abdomen and Pelvis.

CT of the abdomen and pelvis was performed after the administration of contrast material. An axial image (Panel A) shows a curvilinear filling defect within the right renal artery (arrow), a finding consistent with dissection. A coronal image (Panel B) shows a region of hypoenhancement in the right kidney (arrow), a finding suggestive of infarction.

the right kidney. These findings were suggestive of infarction (Fig. 2).

Additional diagnostic tests were performed.

DIFFERENTIAL DIAGNOSIS

Dr. Mark E. Lindsay: This 47-year-old man presented with pain on the left side of the head and neck. His medical history was notable for dyslipidemia and a cerebellar stroke. The development of symptoms similar to those he had noticed before his stroke motivated his quick presentation to the hospital, where imaging studies revealed dissections of the left vertebral artery and right renal artery, with evidence of renal damage.

Identification of the cause of his first stroke had been complicated by other medical events at that time, given that the differential diagnosis of cerebellar stroke is somewhat broad. However, on the current evaluation, there was clear evidence of arterial dissections on imaging, which clarified that the primary medical issue was fragility of medium-sized arteries. With this feature, the differential diagnosis can be narrowed through examination of the associated medical history, laboratory test results, and imaging findings.

MINOR TRAUMA

Could this patient's vertebral artery dissection be associated with his history of chiropractic manipulation? There is a known association between cervical manipulative therapy and cervical artery dissection. Although dissections of the vertebral arteries and of the carotid artery have been reported in patients who have undergone cervical manipulative therapy, the vertebral arteries are particularly vulnerable because of their position in the neck. During cervical manipulative therapy, the V3 segment of the vertebral artery is most susceptible to dissection because rotation and extension stretch the vessel against the atlas or the posterior atlanto-occipital membrane.[1] It is often difficult to determine whether cervical artery dissection has resulted from cervical manipulative therapy on the basis of history alone, because neck pain could be a symptom of the dissection but could also be the reason that the patient initially sought therapy. This patient's neck pain had started before he underwent cervical manipulative therapy, but his stroke symptoms occurred afterward; thus, it is difficult to determine whether cervical manipulative therapy was the cause of his stroke symptoms. However, the identification of renal artery dissection during his subsequent admission suggests that the patient had a more generalized problem of arterial fragility.

VASCULITIS

Vasculitis is characterized by thickening of the vessel wall due to infiltration by inflammatory cells, which weakens the extracellular matrix and increases the risk of arterial stenosis, with dissection occurring in rare cases. Arterial dissection has been associated with many forms of vasculitis — including polyarteritis nodosa, giant-cell arthritis, and Kawasaki's disease — but is a rare event.[2] It occurs at a higher frequency (up to 11%) in patients with Takayasu's arteritis.[3]

Although arterial dissection can occur in the context of these disorders, this patient had no other evidence of vasculitis, such as fever, malaise, weight loss, rash, or generalized, vague pain. In addition, he had no evidence of arterial wall thickening on imaging. Laboratory test results were not suggestive of vasculitis; tests for antinuclear antibodies, antineutrophil cytoplasmic antibodies, and anti-La and anti-Ro antibodies and a rapid plasma reagin test were negative, and the erythrocyte sedimentation rate was only mildly elevated. These findings make vasculitis an unlikely diagnosis in this patient.

ATHEROSCLEROSIS

Could this patient's arterial dissections be an unusual consequence of a common disease? The most common vascular disorder, atherosclerotic vascular disease, can manifest as myocardial infarction, stroke, and peripheral artery disease. Atherosclerosis is a multiregional disease that affects vascular beds, including those in the cervical and abdominal areas. It is characterized by the formation of plaque that consists of fat, cholesterol, calcium deposits, and cellular infiltration in the arterial intima and media; it commonly causes vascular stenosis and regional arterial insufficiency. Vascular dissection can occur when plaque infiltrates and destabilizes the arterial media.

Although this patient had a history of dyslipidemia, several features of his presentation are not consistent with a diagnosis of atherosclerosis. Dissection associated with atherosclerosis typically involves large conduit arteries, such as the aorta, rather than smaller arteries. Also, it typically occurs in older patients and involves vessels with clinically advanced disease. The absence of atherosclerotic plaque on imaging and the relatively young age of this patient make atherosclerosis an unlikely cause of his arterial dissections.

FIBROMUSCULAR DYSPLASIA

Fibromuscular dysplasia is a multiregional arteriopathy that affects a younger patient population than typical atheromatous disease. Fibromuscular dysplasia is characterized by proliferation of intramural cells and deposition of loose, disorganized collagen in the arterial media. Collagen deposition generates fibromuscular ridges; areas of arterial stenosis alternate with areas of arterial dilatation caused by focal cellular loss. Alternating stenoses and dilatations produce the characteristic "beads on a string" appearance of fibromuscular dysplasia on angiography.[4]

The typical manifestations of fibromuscular dysplasia include aneurysm, stenosis, and arterial dissection, and the typical affected territories are consistent with those involved in this patient's presentation. Approximately 20% of patients with fibromuscular dysplasia have arterial dissection; cervical and renal arterial dissections are the most common.[5] However, some aspects of this case do not support a diagnosis of fibromuscular dysplasia. The condition is much more common in women than in men (9:1). Moreover, the characteristic "beads on a string" appearance was not seen on imaging, nor were other findings consistent with fibromuscular dysplasia.

GENETICALLY TRIGGERED ARTERIAL DISEASE

Several aspects of this case are suggestive of genetically triggered arterial disease. Recurrent arterial dissection is a characteristic feature of genetically mediated disease. Furthermore, the patient's age is consistent with the age at which dissection associated with genetically triggered arterial disease typically occurs. The history of major joint dysfunction and easy bruising, with repair of meniscus tears in both knees and repair of shoulder dislocation, may suggest a generalized abnormality of connective tissue.

Several genes have been implicated in arterial dissection. The features of this patient's presentation can help us to determine the gene involved. Mutations in *ACTA2*, *MYH11*, *PRKG1*, and *MYLK*,[6] which are genes that encode regulators of smooth-muscle function, can cause arterial disease, but manifestations are restricted to the cardiovascular system. These mutations would not cause joint or skin dysfunction and would thus be unlikely in this patient. The combination of arterial disease and joint dysfunction is commonly seen in patients with Marfan syndrome, which is caused by mutations in *FBN1*, as well as in patients with Loeys–Dietz syndrome, which is caused by mutations in *TGFBR1*, *TGFBR2*, *TGFB2*, *TGFB3*, *SMAD2*, and *SMAD3*.[7] In addition, arterial dissection and joint dysfunction can be related to deficits in collagen function induced by mutations in the genes that encode procollagen alpha chains. These include mutations in *COL3A1*,[8] which encodes type III collagen, a major structural component of skin, blood vessels, and hollow organs; they less frequently include mutations in *COL1A1*.[9]

Mutations in *COL5A1* and *COL5A2* cause classic Ehlers–Danlos syndrome, a disorder that is infrequently associated with arterial disease, although

COL5A1 has recently been implicated in fibromuscular dysplasia.[10] In this case, there is no mention of a history of skin and joint hyperextensibility, which is the dominant phenotype of classic Ehlers–Danlos syndrome. Arterial disease has also been associated with collagen-modifying enzymes, such as procollagen-lysine, 2-oxoglutarate 5-dioxygenase 1, which is encoded by *PLOD1*, and lysyl oxidase, which is encoded by *LOX*. Mutations in *PLOD1* cause severe joint manifestations and scoliosis, features that are not compatible with this patient's history of manual labor.

Other features of this case can help us to further narrow the list of diagnostic possibilities. Peripheral artery dissection is uncommon in patients with Marfan syndrome but is common in patients with Loeys–Dietz syndrome.[11] That said, thoracic aortic aneurysm is a defining feature of Marfan syndrome[12] and has high penetrance (>90%) in patients with Loeys–Dietz syndrome,[11,13] as well as in patients with *LOX*-associated vascular disease.[14] The findings on echocardiography in this case ruled out aneurysm involving the ascending aorta or aortic root.

In view of all these factors and the overall frequency at which these mutations arise in the population, the most likely genetic diagnosis in this patient, who had recurrent dissection of medium-sized arteries and joint dysfunction in the absence of thoracic aortic aneurysm, is the vascular form of Ehlers–Danlos syndrome, which is caused by mutations in *COL3A1*.

DR. MARK E. LINDSAY'S DIAGNOSIS

Vascular Ehlers–Danlos syndrome.

DIAGNOSTIC TESTING

Dr. Joseph V. Thakuria: A clinical diagnosis of vascular Ehlers–Danlos syndrome can be established when a patient meets one of three major diagnostic criteria: arterial dissection or rupture at a young age, unexplained intestinal rupture, or uterine rupture during pregnancy. Minor criteria that are suggestive of this diagnosis include a predilection for bruising or bruising at atypical sites, spontaneous pneumothorax, small-joint hypermobility, tendon or muscle rupture, early onset of varicose veins, or acrogeria. Patients with a known family history of vascular Ehlers–Danlos syndrome should be offered molecular testing before the onset of symptoms. Given this patient's history of left vertebral artery dissection, right renal artery dissection, and preceding cerebral infarction, we strongly suspected the diagnosis of vascular Ehlers–Danlos syndrome.

Traditional Sanger sequencing for *COL3A1* was performed on a specimen obtained from the patient, and there was a plan to perform a reflex analysis with the use of multiplex ligation-dependent probe amplification (MLPA) and next-generation sequencing panel testing if Sanger sequencing was negative. In more than 95% of patients with vascular Ehlers–Danlos syndrome, a mutation is identified with Sanger sequencing; in approximately 1%, a large deletion is identified by means of MLPA. Biochemical testing of cultured fibroblasts is primarily done to assess the functional effects of splice-site mutations.

In this patient, a mutation at nucleotide position 976 caused a substitution of cytosine for thymine that resulted in a change in coding for arginine to a termination codon (specifically, UGA in this case). Nucleotide triplets within messenger RNA can code for amino acids as well as termination codons ("stop codons"), which signal termination of protein translation and dissociation of ribosomal subunits. Humans have three termination codons in DNA (TAG, TAA, and TGA), with corresponding termination codons in RNA (UAG, UAA, and UGA). This mutation lies in exon 15, but *COL3A1* is 51 exons in length and the protein is more than 1400 amino acids in length. The mutation causes early truncation of the transcription of procollagen. More than 600 pathogenic mutations are known to occur in *COL3A1*, and most are single amino acid substitutions of the glycine in the triple repeat of the triple helical domain of the type III procollagen, with approximately 25% occurring at splice sites.[5] Null *COL3A1* mutations may be associated with a slightly better prognosis than other known pathogenic *COL3A1* mutations.

GENETIC DIAGNOSIS

Vascular Ehlers–Danlos syndrome.

DISCUSSION OF MANAGEMENT

Dr. Thakuria: Vascular Ehlers–Danlos syndrome is a rare autosomal dominant disorder, occurring in 1 in 50,000 to 200,000 patients. Approximately 50% of cases are associated with a de novo (i.e., not inherited) mutation; 2 to 5% of cases are associated with mosaicism, in which the mutation may be confined to one or more specific cell lines. Most of our knowledge about this disorder is derived from two observational studies.[8,15] Of note, patients with null variants, such as this patient, have the longest survival; those with splice-site variants have the shortest survival.[15] Pathogenic mutations in *COL3A1* are known to have 100% penetrance in adults and variable expressivity, even within families.

The presenting sign is vascular dissection, intestinal perforation, or organ rupture in 70% of patients with vascular Ehlers–Danlos syndrome. The median age at death is approximately 50 years and is lower in men (45 years). In one observational cohort, the average age at onset of the first major arterial or gastrointestinal complication was 31 years.[16] Approximately 15% of patients have intestinal perforation, usually in the sigmoid colon, which often requires antibiotic treatment and consideration of surgical intervention. Patients often undergo a partial colectomy with reversal after a few months. Recurrent perforation may warrant colonic resection. Since there are multiple reports of intestinal perforation associated with colonoscopy and there are no specific intestinal findings that predict bowel rupture, colonoscopy for cancer screening is generally not recommended in patients with vascular Ehlers–Danlos syndrome unless there is a perceived high risk and strong family history of cancer. This patient was able to undergo colonoscopy for routine screening after 50 years of age because a gastroenterologist who specialized in vascular Ehlers–Danlos syndrome performed the procedure; virtual colonoscopy and the use of capsular cameras may be options in the future.[16-18] In rare cases of vascular Ehlers–Danlos syndrome, the spleen or liver may rupture.

Pneumothorax can be the presenting sign of vascular Ehlers–Danlos syndrome, and hemopneumothorax and hemoptysis have also been reported.[19-21] There are no specific recommendations regarding pulmonary surveillance in patients with vascular Ehlers–Danlos syndrome.

A carotid–cavernous sinus fistula occurs in 10% of patients, resulting in sudden blurred vision and ocular pain and requiring rapid intervention to preserve the patient's vision. Keratoconus, periodontal disease, and gingival recession are minor criteria. Dermatologic findings include skin fragility and poor wound healing. This patient had undergone an arthroscopic procedure and surgery on his right shoulder and had problems with wound healing and dehiscence. In general, elective surgeries should be avoided in patients with vascular Ehlers–Danlos syndrome; the risks and benefits of any surgery must be weighed carefully, with consideration of the best approach to the procedure (e.g., endovascular or open repair for arterial dissection).[22-25]

Patient and family counseling is essential in the management of vascular Ehlers–Danlos syndrome. We instruct patients to wear a medical alert bracelet in plain view for emergency management staff and to seek immediate medical attention for sudden unexplained pain. Because we learned that this patient had family members with a history of joint hypermobility and dislocations, testing of his extended family was performed, and three affected first-degree relatives were identified.[26]

Dr. Michael R. Jaff: Patients with vascular Ehlers–Danlos syndrome often have understandable difficulty managing the disorder, given its unpredictable nature. Therefore, it is important to have regular discussions with patients about the disorder, the importance of close attention to home monitoring of the blood pressure and heart rate, and the need for emergency medical attention at the onset of new symptoms, regardless of their severity. The use of stress-management tools, biofeedback, psychological counseling, and pharmacotherapy can be helpful in addressing the unpredictable nature of vascular Ehlers–Danlos syndrome.

Lifestyle modifications are commonly recommended, although there is a paucity of scientific evidence regarding the effects of such strategies. Avoidance of activities that pose a risk of trauma to the arteries — for example, riding rollercoasters or skydiving, participating in collision sports, and lifting heavy weights — is sensible. There is controversy regarding the limits of physical activity. It seems unwise to restrict all physical activity, but it seems appropriate to limit exercise intensity according to a target heart rate. Depending on the occupation of the patient, limiting weight lifting to less than 10 kg is recommended.

Aggressive control of the blood pressure and heart rate is a cornerstone of the medical management of vascular Ehlers–Danlos syndrome, and therefore, beta-adrenergic receptor antagonists are commonly used. Regular recording of the blood pressure and heart rate is an important habit for patients to adopt. Unfortunately, there is a paucity of peer-reviewed data suggesting a survival benefit associated with aggressive control. To my knowledge, the only relevant prospective, randomized trial is the BBest trial,[27] which tested the use of celiprolol to reduce arterial events in patients with vascular Ehlers–Danlos syndrome. Celiprolol, which is a third-generation beta-adrenergic receptor antagonist that has not been approved by the Food and Drug Administration, has multiple pharmacologic effects, with activity as a selective $beta_1$ antagonist, a partial $beta_2$ agonist, and a mild $alpha_2$ antagonist. Among the 53 patients with vascular Ehlers–Danlos syndrome who were included in the trial, the frequency of major arterial events and end-organ ischemia in the celiprolol group was one third the frequency in the placebo group. There was no associated reduction in the blood pressure or heart rate.

In patients with vascular Ehlers–Danlos syndrome, surveillance imaging is undertaken to identify early and asymptomatic arterial changes for consideration of preemptive repair. The challenge is that, given the unpredictable nature of the disorder, there is a risk of arterial dissection and rupture even after imaging shows normal arterial segments. However, patients are often reassured by regular surveillance and most clinicians recommend it, especially with the availability of advanced imaging techniques, including duplex ultrasonography and magnetic resonance angiography, which avoid exposures to radiation and iodinated contrast material. It is prudent to perform imaging within 3 months after the index presentation; with control of the blood pressure and heart rate and in the absence of clinical symptoms, it is reasonable to decrease the frequency of imaging to semiannual and ultimately annual intervals.

FOLLOW-UP

Dr. Jaff: This patient has done quite well. At his most recent follow-up visit, more than 10 years after diagnosis, he had no chest, abdominal, or back pain. He has not had any additional strokelike symptoms and can walk unlimited distances without claudication. He monitors his blood pressure at home; the level is routinely 120/80 mm Hg with the use of a regimen that includes metoprolol administered twice daily and amlodipine, hydrochlorothiazide, and losartan administered daily. He avoids strenuous physical activity and limits the amount of weight he must lift at work. On recent surveillance magnetic resonance angiography of the cerebrovascular circulation, the thoracic and abdominal aorta and visceral arteries were normal. Renal function remains normal, and a recent echocardiogram showed no dilatation of the aorta.

FINAL DIAGNOSIS

Vascular Ehlers–Danlos syndrome.

This case was presented at the Medical Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Dr. Pamela W. Schaefer for her assistance with the radiology images.

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